

Multi-Stakeholder Ethical Decision-Making Frameworks for AI Systems

CCDS25-0582 Final-Year Project

Benjamin Oliver Yick (U2120984H)

College of Computing and Data Science, Nanyang Technological University (Singapore)

Academic Year 2025/2026

Supervisor: Dr. Zhang Jiehuang
Examiner: A/P A S Madhukumar



Live Platform Demo: Evaluation Configuration

- MEDF evaluates AI systems against multiple governance frameworks simultaneously.
- Configurable stakeholder weights model different value systems.
- Live at:
<https://ccds25-0582-medf-api.streamlit.app/>.

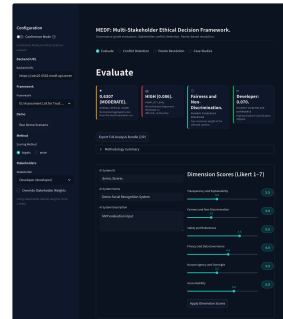


Figure 10.1: MEDF evaluation configuration interface.

Demo: Evaluation Results

- Overall ethical score with risk classification (Low, Medium, High, Critical).
- Radar chart reveals dimensional strengths and weaknesses.
- Per-framework breakdown shows regulatory perspective differences.

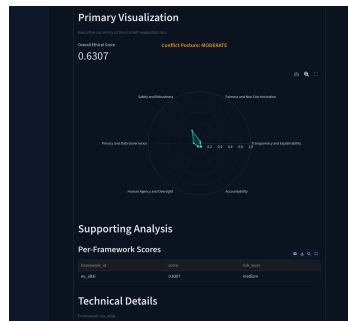


Figure 10.2: MEDF evaluation results visualization.

Demo: Conflict Detection and Pareto Resolution

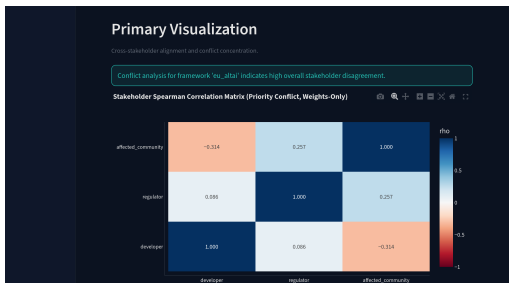


Figure 10.3: Conflict detection heatmap (Spearman ρ).

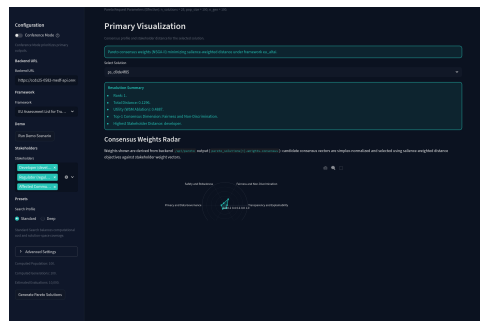


Figure 10.4: Pareto resolution consensus solutions.

Outline

- ① Platform Demo (completed).
- ② Problem Definition and Gap Analysis.
- ③ Methodology, Architecture, and Algorithms.
- ④ Case Study Evaluation.
- ⑤ Limitations, Future Work, and Contributions.

Problem and Motivation

Three Interconnected Sub-Problems in AI governance:

- ① **No standardized metrics.** Ethical concepts remain abstract and context-dependent, making evaluations subjective and incomparable [13].
- ② **Unmanaged stakeholder conflict.** Developers, regulators, and affected communities hold divergent priorities [7, 12].
- ③ **No auditable workflows.** Ethical assessments lack traceability and reproducibility [8].

Real-World Failures: facial recognition bias [4], iTutorGroup age discrimination [20], NHS-DeepMind privacy breach [10]. These patterns are systemic [16, 19].

Gap Analysis: What Existing Tools Miss

Capability	ALTAI	AI Verify	Credo AI	AIF360	Z-Insp.	MEDF
Cross-framework comparison	—	—	—	—	—	✓
Stakeholder weighting	—	—	—	—	—	✓
Conflict detection	—	—	—	—	—	✓
Pareto consensus	—	—	—	—	—	✓
Quantitative scoring	—	✓	✓	✓	—	✓
Multi-dimensional viz.	—	✓	✓	✓	—	✓

Source: Comparative analysis from Chapter 12.3, based on Morley et al. [14] and Zicari et al. [22].

Methodology: Design Science Research

MEDF follows the Design Science Research Methodology (DSRM) by Peffers et al. [18], providing academic legitimacy beyond pure software engineering.

- ① **Problem Identification:** Principles-to-practice gap in AI ethics (Chapters 2–3).
- ② **Objective Definition:** Quantifiable, multi-stakeholder evaluation (Chapter 3).
- ③ **Design and Development:** Architecture, algorithms, and implementation (Chapters 6–9).
- ④ **Demonstration and Evaluation:** Three real-world case studies (Chapters 10–11).

Framework Harmonization: Six Unified Dimensions

Three governance frameworks harmonized into one evaluation model.

Source Frameworks:

- EU ALTAI [1, 2].
- NIST AI RMF [15].
- Singapore MGAF [17].

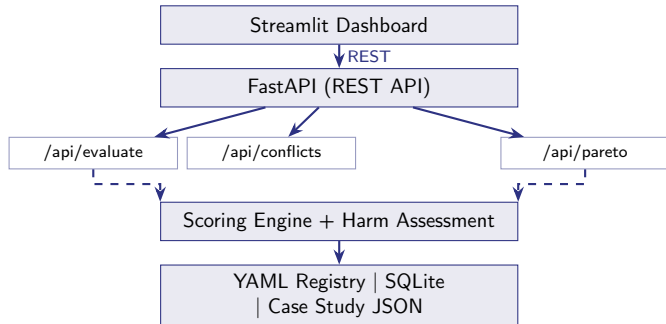
Six Unified Dimensions:

- ① Transparency and Explainability.
- ② Fairness and Non-discrimination.
- ③ Safety and Robustness.
- ④ Privacy and Data Governance.
- ⑤ Human Agency and Oversight.
- ⑥ Accountability.

Derived from overlapping themes across all three frameworks, informed by Jobin et al. [11], Floridi et al. [6], and Zhang et al. [21]. Stored as version-controlled YAML files.

System Architecture: Three-Tier Design

- **Presentation:** Streamlit dashboard with interactive sliders and radar charts.
- **API/Orchestration Layer:** FastAPI with Pydantic validation, routing, and workflow coordination.
- **Data Layer:** SQLite, YAML registry, and JSON case studies.



Algorithm: TOPSIS Scoring Engine

Product Pooling combines stakeholder and framework weights [9, 3]:

$$w_{\text{eff}} \propto w_s \odot w_f$$

TOPSIS Closeness Coefficient (distance to ideal vs. anti-ideal):

$$C = \frac{d^-}{d^+ + d^-} \in [0, 1]$$

- Single-alternative evaluation uses synthetic best/worst anchors to keep C well-defined.
- Risk classification: $C \geq 0.80$ Low, $0.60\text{--}0.80$ Medium, $0.40\text{--}0.60$ High, < 0.40 Critical.

Algorithm: Conflict Detection (Spearman ρ)

MEDF computes two types of stakeholder correlation.

- ① **Priority Conflict (Weights-Only):** Spearman ρ between raw stakeholder weight vectors. Measures general agreement on dimension importance.
- ② **System-Salience Conflict (Contribution-Based):** Spearman ρ between contribution vectors (effective weights $\times$ dimension scores). Measures agreement for a specific AI system.

Critical Insight: The contribution-based metric produces different conflict levels across case studies, even with identical stakeholder weights. This captures case-specific dynamics.

Algorithm: Pareto Consensus and Harm Assessment

NSGA-II objective function with salience weighting [5]:

$$f_k(w_c) = \sum_i s_i \cdot |w_{c,i} - w_{k,i}|$$

Harm Score combines system performance and governance disagreement:

$$h = 0.7 \times (1 - s_{\text{norm}}) + 0.3 \times \bar{d}$$

- The 0.7/0.3 split reflects a design decision: system performance dominates risk classification, with stakeholder disagreement as an aggravating factor.

Case Study 1: Metropolitan Police Live Facial Recognition

Baseline Scores: Fairness 1.5, Privacy 1.5, Safety 5.0 (1–7 Likert scale).

Results (All Frameworks):

- Overall: **0.3866** (Critical).
- Developer: 0.4826.
- Affected Community: 0.3063.

Conflict: Low (all $\rho > 0.89$). The extreme ethical profile creates clear ranking agreement.

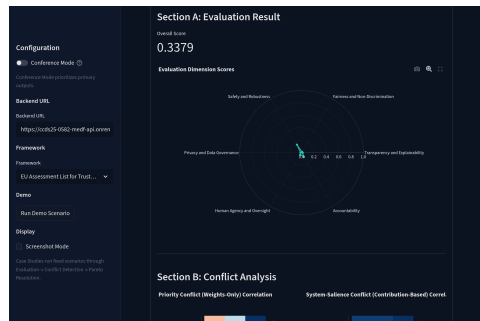


Figure 10.6: Case Study 1 evaluation results (overall score 0.3379).

Case Study 2: iTutorGroup Automated Hiring

System Description: Automated rejection of applicants over age 55 [20]. Deliberate, systemic age discrimination.

MEDF Evaluation Results (All Frameworks):

- Overall: **0.3477** (Critical risk). A single catastrophic ethical flaw renders the system unacceptable from every stakeholder perspective.
- Conflict: Moderate ($\rho = 0.54$ Developer-Regulator). The ambiguous ethical profile exposes genuine disagreement.
- The Regulator scores higher (0.4004) due to Accountability emphasis, while the Affected Community scores lowest (0.3005).

Case Study 3: Royal Free NHS-DeepMind Streams

System Description: Clinical AKI detection app. Life-saving benefit, but 1.6M patient records accessed without consent [10].

MEDF Evaluation Results (All Frameworks):

- Overall: **0.6003** (Medium risk). The genuine clinical benefit (Safety 6.0) partially offsets the Privacy failure (3.5).
- Conflict: High ($\rho = -0.14$ Developer-Regulator). The balanced ethical profile exposes real stakeholder tension.
- This is the most genuinely contentious case: the Developer values clinical utility while the Regulator prioritizes data governance.

Cross-Case Comparative Analysis

Case Study	Overall	Risk	Min ρ	Conflict
Facial Recognition	0.3866	Critical	0.89	Low
iTutorGroup Hiring	0.3477	Critical	0.54	Moderate
NHS-DeepMind Streams	0.6003	Medium	-0.14	High

Critical Finding: Contribution-based conflict analysis differentiates genuinely contentious cases from clear ethical violations, even when using identical default stakeholder weights.

Limitations

- ① **Input Subjectivity.** Baseline dimension scores and stakeholder weights are user-provided. No built-in mechanism verifies their accuracy against ground truth.
- ② **Linearity and Independence Assumptions.** TOPSIS and WSM assume dimensions are independent. In reality, improving transparency can affect accountability.
- ③ **No Primary User Research.** The project scope excluded user surveys, interviews, and statistical user studies.
- ④ **Router-Level Technical Debt.** Conflict analysis logic remains in the API router layer rather than an extracted standalone engine module.

Future Work

- ① **NLP-Based Automated Scoring.** Generate baseline dimension scores from system documentation and audit reports, reducing reliance on subjective manual input.
- ② **CI/CD Pipeline Integration.** Embed MEDF evaluations as automated ethical checks in the software development workflow.
- ③ **Empirical User Studies.** Validate usability and perceived value with AI practitioners, ethicists, and policymakers.
- ④ **Expanded Framework Library.** Add IEEE Ethically Aligned Design, OECD AI Principles, and sector-specific regulations.

Contributions

- ① **Unified Six-Dimension Model** harmonizing EU ALTAI, NIST AI RMF, and Singapore MGAF into a single coherent evaluation framework.
- ② **TOPSIS/WSM Scoring Engine** with product pooling for integrated stakeholder-framework weight computation.
- ③ **Contribution-Based Conflict Detection** via Spearman rank correlation, producing case-sensitive stakeholder disagreement analysis.
- ④ **NSGA-II Pareto Consensus Generation** for multi-stakeholder mediation with salience-weighted objective functions.
- ⑤ **Open-Source, Reproducible Platform** with 50+ automated tests, public deployment, and complete documentation.

References I

- [1] AI HLEG. *Ethics Guidelines for Trustworthy AI*. Tech. rep. Brussels, Belgium: European Commission, 2019.
- [2] AI HLEG. *The Assessment List for Trustworthy Artificial Intelligence (ALTAI) for Self-Assessment*. Tech. rep. Brussels, Belgium: European Commission, 2020.
- [3] Majid Behzadian et al. “A State-of-the-Art Survey of TOPSIS Applications”. In: *Expert Systems with Applications* 39.17 (2012), pp. 13051–13069. DOI: 10.1016/j.eswa.2012.05.056.
- [4] Joy Buolamwini and Timnit Gebru. “Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification”. In: *Proceedings of the Conference on Fairness, Accountability, and Transparency*. Vol. 81. Proceedings of Machine Learning Research. 2018, pp. 77–91.

References II

- [5] Kalyanmoy Deb et al. “A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II”. In: *IEEE Transactions on Evolutionary Computation* 6.2 (2002), pp. 182–197. DOI: 10.1109/4235.996017.
- [6] Luciano Floridi, Josh Cowls, et al. “AI4People: An Ethical Framework for a Good AI Society”. In: *Minds and Machines* 28.4 (2018), pp. 689–707. DOI: 10.1007/s11023-018-9482-5.
- [7] R. Edward Freeman. *Strategic Management: A Stakeholder Approach*. Boston, MA, USA: Pitman, 1984. ISBN: 978-0273019138.
- [8] Thilo Hagendorff. “The Ethics of AI Ethics: An Evaluation of Guidelines”. In: *Minds and Machines* 30.1 (2020), pp. 99–120. DOI: 10.1007/s11023-020-09517-8.

References III

- [9] Ching-Lai Hwang and Kwangsun Yoon. *Multiple Attribute Decision Making: Methods and Applications*. Berlin, Germany: Springer, 1981. DOI: 10.1007/978-3-642-48318-9.
- [10] ICO. *Royal Free: Google DeepMind Trial Failed to Comply with Data Protection Law*. Tech. rep. Wilmslow, UK: Information Commissioner's Office, 2017.
- [11] Anna Jobin, Marcello Ienca, and Effy Vayena. "The Global Landscape of AI Ethics Guidelines". In: *Nature Machine Intelligence* 1.9 (2019), pp. 389–399. DOI: 10.1038/s42256-019-0088-2.
- [12] Ronald K. Mitchell, Bradley R. Agle, and Donna J. Wood. "Toward a Theory of Stakeholder Identification and Salience: Defining the Principle of Who and What Really Counts". In: *Academy of Management Review* 22.4 (1997), pp. 853–886. DOI: 10.5465/amr.1997.9711022105.

References IV

- [13] Brent Mittelstadt. “Principles Alone Cannot Guarantee Ethical AI”. In: *Nature Machine Intelligence* 1.11 (2019), pp. 501–507. DOI: 10.1038/s42256-019-0114-4.
- [14] Jessica Morley et al. “From What to How: An Initial Review of Publicly Available AI Ethics Tools, Methods and Research to Translate Principles into Practices”. In: *Science and Engineering Ethics* 26.4 (2020), pp. 2141–2168. DOI: 10.1007/s11948-019-00165-5.
- [15] NIST. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. Tech. rep. NIST AI 100-1. Gaithersburg, MD, USA: National Institute of Standards and Technology, 2023. DOI: 10.6028/NIST.AI.100-1.

References V

- [16] Cathy O'Neil. *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy*. New York, NY, USA: Crown Publishing Group, 2016. ISBN: 978-0553418811.
- [17] PDPC. *Model Artificial Intelligence Governance Framework*. Tech. rep. Singapore: Personal Data Protection Commission, 2020.
- [18] Ken Peffers et al. "A Design Science Research Methodology for Information Systems Research". In: *Journal of Management Information Systems* 24.3 (2007), pp. 45–77. DOI: 10.2753/MIS0742-1222240302.
- [19] Inioluwa Deborah Raji et al. "Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing". In: *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT*)*. 2020, pp. 33–44. DOI: 10.1145/3351095.3372873.

References VI

- [20] U.S. Equal Employment Opportunity Commission. *EEOC Sues iTutorGroup for Age Discrimination*. Press Release. May 2022.
- [21] Jiehuang Zhang, Yixing Shu, and Han Yu. “Fairness in Design: A Framework for Facilitating Ethical Artificial Intelligence Designs”. In: *International Journal of Crowd Science* 7.1 (2023), pp. 32–39. DOI: 10.26599/IJCS.2022.9100033.
- [22] Roberto V. Zicari, John Brodersen, James Brusseau, et al. “Z-Inspection: A Process to Assess Trustworthy AI”. In: *IEEE Transactions on Technology and Society* 2.2 (2021), pp. 83–97. DOI: 10.1109/TTS.2021.3066209.

Backup: Stakeholder Weight Profiles

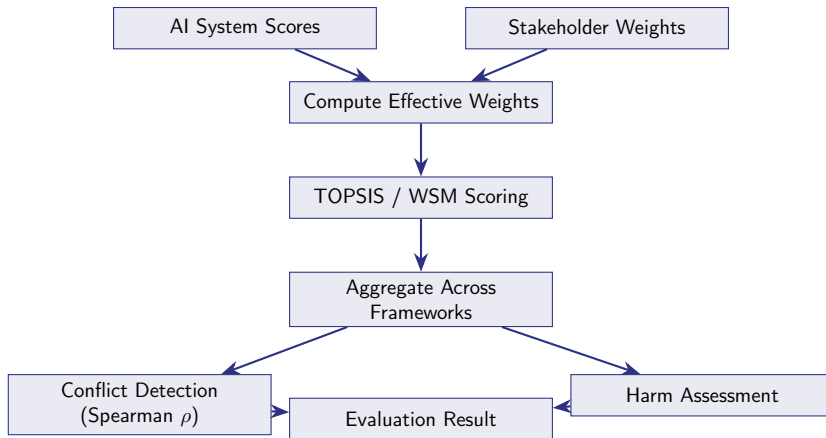
Dimension	Developer	Regulator	Affected Community
Transparency	0.10	0.20	0.10
Fairness	0.15	0.20	0.30
Safety	0.30	0.10	0.10
Privacy	0.15	0.15	0.15
Human Oversight	0.15	0.10	0.20
Accountability	0.15	0.25	0.15
Total	1.00	1.00	1.00

Backup: Framework Weight Vectors

Dimension	EU	ALTAI	NIST	RMF	SG	MGAF
Transparency	0.12		0.20		0.22	
Fairness	0.14		0.20		0.12	
Safety	0.22		0.24		0.16	
Privacy	0.20		0.12		0.12	
Human Oversight	0.18		0.12		0.18	
Accountability	0.14		0.12		0.20	
Total	1.00		1.00		1.00	

Weights derived from the structural emphasis of each framework (requirement and principle counts).

Backup: End-to-End Evaluation Pipeline



Backup: Module Dependency Diagram

